

LESSON 6-5 • SESSION 2 • PRACTICE • FORM A

Evaluating Algebraic Expressions

Learning target – I can evaluate an algebraic expression by substituting a value for the variable. · 6.EE.A.2c

REMEMBER

- To **evaluate**, substitute the value, then use the **order of operations**.
- $4x$ means $4 \times x$. When $x = 3$, $4x = 12$, not 43.
- Powers come before multiplying: $2n^2$ with $n = 5$ is $2 \times 25 = 50$.

WORKED EXAMPLEEvaluate $2n^2 + 3$ when $n = 5$.**Step 1** Substitute: $2 \times 5^2 + 3$.**Step 2** Power first: $5^2 = 25$, so $2 \times 25 + 3$.**Step 3** Multiply, then add: $50 + 3 = 53$.**Answer 53****WORDS** substitute (*sustituir*) · evaluate (*evaluar*) · variable (*variable*)**PRACTICE***show your work · label units*

1. Evaluate each expression when
- $b = 7$
- .

a. $6b - 9 =$ _____

b. $5b =$ _____

c. $b + 18 =$ _____

d. $50 - 4b =$ _____

e. $b^2 - 9 =$ _____

f. $2(b + 3) =$ _____

- 2.
- Parking Garage.**
- A garage charges
- $4h + 5$
- dollars for
- h
- hours of parking.

a. Cost for 3 hours: _____ dollars

b. Cost for 8 hours: _____ dollars

c. How much more is 8 hours than 3 hours? _____ dollars

d. What does the 5 do to every cost? _____

- 3.
- Bike Rental.**
- The expression
- $9r + 14$
- gives the cost in dollars for
- r
- hours. Find the cost for 4 hours.

Hint Multiply first, then add 14.*Show your work.***Answer** _____ dollars

- 4.
- Power Inside.**
- Evaluate
- $5k^2 - 10$
- when
- $k = 3$
- .

Hint Square 3 before you multiply by 5.*Show your work.***Value** _____

- 5.
- Yara's Work.**
- Yara evaluated
- $4 + 2p$
- when
- $p = 5$
- and got 30. Justify or correct her work.

Hint After you substitute, which operation comes first?*Show your work.***Value** _____

6. **Draining Tank.** The expression $25 - 3d$ gives the liters of water left in a tank after d days.

Hint Substitute each value on its own.

- a. Liters left when $d = 2$: _____ L
 b. Liters left when $d = 8$: _____ L
 c. What happens as d grows? _____

Show your work.

7. **Team Points.** A league uses $3w + 2t$ for points, where w is wins and t is ties.

Hint Substitute both values, then add the products.

- a. Points for 5 wins and 4 ties: _____ points
 b. Points for 8 wins and 0 ties: _____ points

Show your work.

8. **Phone Plan.** A plan costs $12g + 20$ dollars, where g is the number of gigabytes used.

- a. Cost for 3 gigabytes: _____ dollars
 b. Cost for 6 gigabytes: _____ dollars
 c. The gigabytes doubled. Why didn't the cost double? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Hugo's Work.** Hugo wrote "15 less than a number x " as $15 - x$. Explain his mistake. Then write the correct expression and evaluate it when $x = 22$.

TEST PRACTICE

choose the best answer

10. Find the value of $2n + 7$ when $n = 4$.

A 15 **B** 31 **C** 13 **D** 24

11. Substituting means replacing the variable with a –

A number **B** letter **C** sign **D** unit

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

19 U	35 A	215 Z	92 E	23 A	50 V	56 T	37 E	30 Q	14 L
#2b	#3	#4	#5	#6a	#7a	#8a	#8b		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____